Football Tactical Intelligence Platform

Android App + Scalable Backend — Product and Engineering Plan

Status: Product blueprint / implementation plan
Platform: Android-first, with a backend designed to support a future web/tablet analyst workspace
Primary users: Football fans first; tactical enthusiasts, coaches, scouts and analysts as progressively deeper user groups
Working product name: Touchline (placeholder; validate naming and trademarks before launch)

1. Product vision

Build a **fan-first football tactical intelligence app**. Its default experience helps an ordinary supporter understand what is happening beyond the scoreline: why a team created chances, why a press worked, what changed after half-time and how a player's role affects the match. Tactical enthusiasts can investigate the evidence, while coaches, scouts and analysts can access the same data through deeper professional workflows. The product must never feel like a coaching tool that casual fans have to learn before they can use it.

The central product promise is:

> Understand why a team plays the way it does, what is happening on the pitch, and how tactical decisions shape the match.

Audience and jobs to be done

Audience	What they come to understand or do	Default experience
Curious / casual fan (primary)	Understand why a team won, lost, dominated, struggled or changed approach	Simple match stories, visual explanations and clear takeaways
Engaged football fan	Explore pressing, build-up, space, transitions and player roles	Interactive tactical breakdowns, team-style profiles and comparisons
Tactical enthusiast	Investigate detailed match phases, metrics, patterns and player behaviour	Advanced filters, pitch overlays, player-role analysis and metric definitions
Coach / scout / analyst	Review performance, assess opponents and evaluate tactical execution	Optional advanced workspace, game models and evidence-linked reports

These are not separate apps or separate databases. They are progressively deeper ways of using the same football intelligence platform.

Product principles

1. **Accessible first, deep on demand:** explain the key takeaway in ordinary language, then let users open the supporting metrics and rawer detail.
1. **Evidence before opinion:** every insight should link to source metrics, match events, tracking frames or authorized video timestamps.
1. **Explain the jargon:** terms such as xG, PPDA, half-space, rest defence and line-breaking pass need short definitions, diagrams and examples.
1. **Never pretend data exists:** clearly indicate whether a match has event data, freeze-frame data, continuous tracking or video.
1. **Separate fact from interpretation:** distinguish observed events, calculated metrics, model estimates and AI-generated hypotheses.
1. **Useful free tier:** a fan should be able to learn and explore meaningful tactical information without paying.
1. **No artificial paywalls in core workflows:** don't interrupt a match explanation or tactical exploration to sell a subscription.
1. **Global football discovery:** users can follow real-world clubs, competitions and players from any competition covered by the selected providers.
1. **Fan-first product hierarchy:** the first screen and default match experience are designed for ordinary fans; coach/analyst tools are optional deeper layers, not the product's identity.

1. **Incremental architecture:** begin with a modular monolith and separate worker processes; split services only when scale or independent deployment justifies it.

Experience design rule: three levels of depth

1. **Quick view — What happened?** A concise takeaway, simple visual and a short explanation.
2. **Explore — Why did it happen?** Interactive pitch maps, player roles, phases, comparisons and supporting metrics.
3. **Deep analysis — Show me the detail.** Tracking overlays, custom filters, metric definitions, tactical patterns and analyst tools where the source data supports them.

The interface should make the next level discoverable without forcing a new user to learn the language of professional analysis first.

Core product decisions

Decision	Product choice
Primary audience	Everyday football fans who want to understand tactics; experts are a deeper audience, not the default persona
Football coverage	Search and follow any club, national team, player or competition in the provider-backed catalogue, across supported leagues and countries
Team model	Real-world teams are shared global football records; users follow them. Do not create private copies of clubs
Account model	One personal account, saved items and simple ownership checks; no organization hierarchy or staff RBAC in v1
Experience model	Quick → Explore → Advanced; users can change depth at any time
Data strategy	Event data first; full player tracking and video only where a provider licence and coverage support them
Monetization	Free is genuinely useful; one optional Pro subscription sells greater depth, history and workflow capacity
Architecture	Kotlin Android app + NestJS modular backend + PostgreSQL + ClickHouse + Redis/BullMQ workers + object storage

Important: Audience type is not an account role or subscription tier. A fan may use advanced analysis, and a coach may prefer the quick explanation. The user chooses the depth; the product does not stereotype the user.

2. Accounts, followed football and simple access control

Keep account and permission logic deliberately simple. The normal user is following existing real-world football entities, not creating an organization or managing staff.

User model

1. Every person has one personal account.
1. The default account experience is to follow existing football clubs, players and competitions; it is not to create or administer a team.
1. Users can follow multiple real-world clubs, competitions and players.
1. Real-world clubs, competitions, seasons, matches and player records live in a shared canonical football database. Do not create a private duplicate of Arsenal, Barcelona or another club for every user.
1. Each account has private saved matches, notes, bookmarks, collections, preferences and analysis history.
1. The normal use case is following existing professional or national teams from any supported league. A user-created **custom team workspace** for a local, amateur or academy side is a later optional feature, not part of the initial fan product.
1. There is no organization hierarchy, membership system, staff-seat model or general-purpose RBAC in v1.

```
Shared football database
■■■ Competitions and seasons
■■■ Real-world clubs
■■■ Players and squads
■■■ Fixtures and matches
■■■ Events and statistics
```

```
■■■ Tracking assets (only where licensed)

Personal account
■■■ Followed clubs / competitions / players
■■■ Saved matches and comparisons
■■■ Bookmarks and tactical notes
■■■ Preferences and learning progress
■■■ Subscription and entitlements
■■■ Optional custom team workspace (later)
```

Access-control rules for v1

1. Authenticate the user.
2. Read `user_id` from the verified backend session, never from a client-supplied owner field.
3. For private notes, saved lists, reports, uploads and preferences, verify the resource's `owner_user_id` equals the authenticated `user_id`.
4. Shared football data is read-only from the user's perspective and governed by the provider's licence and the app's data access rules.
5. If custom team workspaces are introduced, keep them owner-only initially and apply the same simple ownership check.
6. Subscription entitlements determine feature and usage access; they do not determine data ownership or permissions.

Defer until there is real demand

Team sharing, assistant-coach accounts, player logins, invitations, shared workspaces, staff seats, organization-level billing and configurable permissions should not be part of v1. If collaboration becomes necessary, begin with only `owner`, `editor` and `viewer` on a specific custom workspace; do not build a configurable permission engine prematurely.

3. Application information architecture and navigation

The app should feel like a football discovery and analysis product first. Do not make a new user create a squad, configure a game model or understand coaching terminology before seeing value.

First-time user journey

1. Welcome screen explains the promise: understand football beyond scores and basic statistics.
2. User signs in with Google, passkey or email; allow a limited preview before sign-up where practical.
3. Ask the user to select favourite clubs and competitions, with search across supported leagues.
4. Offer a light preference step: favourite clubs/competitions and interests such as match explanations, tactics, players or learning football concepts. Optionally ask whether the user prefers simple explanations or detailed statistics. Do not ask users to label themselves as a coach, fan or analyst; this only personalises the default presentation depth, not access to the product.
5. Show a personalised home feed with upcoming matches, recent results and one or two understandable tactical insights.
6. Open a match and present a simple explanation first; advanced details are available through “Explore this” and “Deep analysis”.
7. Let users save a match, follow a player or bookmark an insight.

Do not force users to select only a single team. A fan may follow many clubs and competitions.

Users should be able to search for and follow **any real-world club, national team, player or competition present in the supported provider catalogue**, regardless of league or country. The catalogue is global and shared across users. Coverage varies by provider, competition, season and data type; show a clear coverage label before users expect advanced tactical features. Do not imply that every team has continuous tracking or video.

Main navigation

Keep bottom navigation to a small number of destinations:

1. **Home** — followed-club activity, upcoming matches, recent tactical stories and recommended explainers.
2. **Explore** — search and browse competitions, clubs, players, matches and tactical topics.
3. **Matches** — followed-club fixtures, match centre, timelines and comparisons.

4. **Learn / Tactics** — visual tactical explainers, glossary, tactical concepts and deeper analysis tools.
5. **Profile** — following list, saved items, preferences, subscription and settings.

Team and player pages should be reached through search, following lists and match pages. Professional workflows such as custom game models, opponent reports and advanced exports should appear contextually or inside an optional Advanced Analysis area. They must not dominate the main navigation.

Primary screens

Screen	Main purpose	Important actions
Welcome / sign-in	Explain the value and provide fast account entry	Google, passkey, email; optional preview
Onboarding	Personalise the football experience	Follow clubs, competitions and players; choose interests
Home	Show relevant football and tactical insights	Open match, read explanation, follow team
Explore / search	Discover football entities and concepts	Search any supported league, club, player, match or tactic
Competition page	Understand a league or tournament	Fixtures, standings, form and tactical trends where available
Club page	Understand a team's identity and performance	Formation tendencies, style, players, matches and tactical trends
Match centre	Understand a match at a glance	Timeline, stats, xG, phases, lineups and explanations
Tactical pitch	Explore spatial and phase-based patterns	Switch teams, periods, players, overlays and depth level
Player profile	Understand a player's role and contribution	Role explanation, trends, comparisons and clips where licensed
Tactical concept page	Learn a concept in context	Plain-language explanation, diagram, match examples and glossary
Advanced analysis	Investigate detailed tactical questions	Filters, tracking overlays, pattern analysis and custom reports
Saved items	Return to useful football content	Saved matches, players, insights and notes
Subscription	Explain value and manage plan	Plans, billing status, restore and manage subscription

UX rules

1. Default to plain-language summaries; expose technical labels with definitions on tap.
1. Every insight should answer: **What happened? Why does it matter? What evidence supports it?**
1. Use progressive disclosure; never put dozens of metrics on the first screen.
1. Provide an optional “Explain this” action for every advanced metric or tactical term.
1. Use short visual explainers rather than walls of text.
1. Every chart needs a title, units, definition, selected time range and data-source/freshness indicator.
1. Never use colour alone to communicate a status; combine colour with text, icons or patterns.
1. Maintain readable contrast, scalable typography and minimum comfortable touch areas.
1. Use skeletons for loading; distinguish loading, empty, unavailable, stale and error states.
1. Offer useful empty states with a clear next step (e.g. “Follow a club” or “Explore a match”).
1. Save filters and the user's last-used team/match context.
1. Make destructive actions reversible where possible; confirm account deletion and deletion of custom workspaces.
1. Support dynamic text sizing, TalkBack labels, keyboard navigation where applicable, reduced motion and light/dark themes.
1. Do not force landscape orientation globally; use landscape as an optional tactical/video workspace mode.

- Keep the match explanation and tactical pitch ad-free.

4. Product feature catalogue

The feature set should be implemented in layers based on available data. A feature must not be marked as supported if the required source data is missing.

4.1 Match centre and timeline

- Fixtures, results, lineups, substitutions, formations and match events.
- Match timeline with goals, cards, shots, substitutions, tactical notes and saved clips.
- Team/player statistics, possession, passes, shots, corners and xG when supplied/licensed.
- First-half / second-half / custom time-window comparison.
- Score-state comparison: leading, drawing and trailing.
- Match-to-match comparison across a selected date range.
- Data freshness and coverage indicators.
- Event search and filters (e.g. turnovers in defensive third, progressive passes, shots after counter-attacks).

4.2 Team tactical intelligence

Requires tracking data for high-confidence spatial analysis. Event data alone can support only a subset of these metrics.

- Dynamic formation detection by possession phase.
- In-possession and out-of-possession team shape.
- Team width, length, occupied area and centroid.
- Defensive line height and goalkeeper position.
- Distances between defensive, midfield and forward lines.
- Team compactness and line-to-line spacing.
- Rest-defence structure while attacking.
- Space occupation and unoccupied-space maps.
- Formation stability and structural changes after substitutions.
- Positional rotations and player role exchanges.
- Shape comparison by opponent, score state and match period.

4.3 Build-up and possession

- Build-up success and progression rate.
- First-phase build-up patterns and goalkeeper involvement.
- Press resistance and possession retention under pressure.
- Line-breaking passes and carries.
- Progressive passes/carries and progression zones.
- Third-player / third-man combinations.
- Overloads and numerical superiority in zones.
- Switches of play and weak-side exploitation.
- Receiving between lines and access to central zones.
- Possession chains leading to shots or penalty-area entries.
- Turnover location and build-up failure classification.
- Possession value: xT, EPV or VAEP-style metrics, with methodology documented.

4.4 Pressing and defending

- [1] PPDA, with a clearly documented definition and provider-specific comparability caveat.
- [1] Pressing frequency, location, intensity and success.
- [1] High turnovers and shots created after high turnovers.
- [1] Counterpress regain rate within configurable time windows.
- [1] Pressing triggers and opponent press-escape patterns.
- [1] Pressing traps and forced-direction patterns.
- [1] Cover-shadow and passing-lane blocking estimates (tracking-dependent).
- [1] Defensive block height, width, depth and compactness.
- [1] Opponent progression allowed by zone.
- [1] Defensive line coordination and offside-line behaviour where data supports it.
- [1] Recovery shape after a line is broken.

4.5 Attacking threat and chance creation

- [1] xG / non-penalty xG / xG per shot when a licensed provider supplies the model or the app has a validated model.
- [1] Expected assists and shot assists where supplied or computed from licensed events.
- [1] Expected Threat (xT) and possession value.
- [1] Box entries, penalty-area touches and cutbacks.
- [1] Chance-creation chains and sequences.
- [1] Shot location, angle, body part, pressure and situation.
- [1] Cross quality and delivery-zone analysis.
- [1] Attacking overloads and weak-side access.
- [1] Chance creation patterns by player combination and pitch zone.
- [1] Actual goals compared with expected output, with uncertainty and sample-size context.

4.6 Transition analysis

- [1] Time from loss of possession to first counterpress action.
- [1] Regain probability and successful counterpress rate in 3/5/8-second windows.
- [1] Time from regaining possession to entering the final third or penalty area.
- [1] Counterattack speed, distance, player involvement and outcome.
- [1] Defensive recovery time and shape restoration.
- [1] Space exposed behind the ball during attacking phases.
- [1] Opponent counterattack patterns and transition vulnerability.

4.7 Player tactical intelligence

- [1] Role-based player evaluation rather than one universal rating.
- [1] Role profiles: holding midfielder, deep-lying playmaker, box-to-box midfielder, ball-winner, advanced playmaker, inverted fullback, overlapping fullback, winger, target forward, pressing forward and others.
- [1] Off-ball runs and movement maps.
- [1] Receives between lines and access to valuable zones.
- [1] Progressive actions, line breaks and possession value.
- [1] Pressing contribution and defensive coverage.
- [1] Space creation for teammates (tracking-dependent).

- Passing decisions against available options (tracking-dependent).
- Physical load: distance, high-speed running, sprinting, acceleration/deceleration (source and thresholds must be stated).
- Percentile comparisons against similar roles, competition and minutes played.
- Player trend over last N matches and season.
- Player similarity search with explainable attributes.

4.8 Opposition scouting

- Opponent build-up structures and preferred progression channels.
- Pressing height, triggers and escape routes.
- Defensive line and block structure.
- Weaknesses by zone and phase.
- Attacking patterns and recurring combinations.
- Transition vulnerabilities and counterattack tendencies.
- Set-piece routines and defensive schemes.
- Key player role profiles and matchups.
- Performance split when leading, drawing and trailing.
- Auto-generated opponent report with source-linked evidence and video timestamps.

4.9 Set-piece intelligence

- Corner routines classified by delivery zone and movement pattern.
- First-contact success and second-ball recovery.
- Near-post, central and far-post targeting.
- Short-corner and long-corner effectiveness.
- Free-kick delivery and routine patterns.
- Defensive zonal/man-marking scheme classification where video/tracking supports it.
- Aerial duel profiles and matchup analysis.
- Set-piece xG and goals above/below expected.
- Searchable routine library with opponent, match and outcome filters.

4.10 Advanced differentiators

These capabilities are optional deeper layers. They should not be prerequisites for understanding a match, and they should not define the app's first-run experience.

A. Personal tactical model (advanced coach / analyst feature)

Allow a coach or analyst to define desired team principles for:

- Build-up shape and preferred numerical structures.
- Progression zones and passing preferences.
- Pressing height, triggers and direction.
- Defensive block, line height and compactness.
- Transition response after winning/losing possession.
- Player responsibilities by role.

Measure actual match behaviour against the model. Let the owner customise weights and acceptable ranges. Explain why each principle was marked as achieved, partially achieved or missed, with links to the supporting match sequences. Keep this feature optional and out of the casual fan's default journey.

B. Automatic tactical pattern discovery

Detect and cluster repeated sequences such as:

1. Fullback/winger rotations.
1. Third-player runs.
1. Central overloads.
1. Switches followed by weak-side entries.
1. Repeated press triggers.
1. Opponent escape patterns.
1. Defensive gaps after a specific player advances.

Show example matches and clips for every discovered pattern. Analysts must be able to confirm, rename, dismiss and annotate the pattern.

C. Ask about this match (AI football explainer)

Allow natural-language questions such as “Why did they struggle to build out in the second half?”, “Why did this winger keep finding space?” or “What changed after the substitution?” Coaches and analysts can ask more technical questions about their own analysis.

The AI must:

1. Resolve the selected team, match, opponent and time period.
2. Query structured metrics and event/tracking evidence.
3. Retrieve relevant sequences and video timestamps if licensed.
4. Return a concise answer with supporting data and links to evidence.
5. Distinguish observations from hypotheses and recommendations.
6. State when the data is insufficient.
7. Never invent a player position, event or video clip.

Use the LLM for synthesis and interaction, not as the source of truth for numeric metrics.

D. Tactical what-if workspace (advanced)

Let fans explore simple hypothetical shape changes and let coaches/analysts compare scenarios such as:

1. 4-3-3 versus 3-4-2-1.
1. Higher versus deeper defensive line.
1. Different pressing triggers.
1. Alternative player roles.
1. Different build-up structures.

Start with visual scenario editing and historical analog retrieval. Later add a validated simulation model. Label outputs as hypothetical scenario estimates, not guaranteed match outcomes.

E. Video-synchronized analysis

1. Link events and tracking frames to authorized match footage.
1. Create timestamped clips and playlists.
1. Annotate movements, lanes and player responsibilities.
1. Save personal notes and, where sharing rights exist, create role-specific review playlists.
1. Sync pitch position and video timecode.
1. Export only where video contracts permit it.

Do not scrape or redistribute broadcast footage. Video rights, recording rights and clip-export rights must be explicitly licensed or supplied by the club/user under suitable terms.

4.11 Fan-first tactical explainability and learning layer

This layer is a core product capability, not a separate beginner mode or a collection of articles. It makes the advanced analytics understandable without diluting the underlying data.

Insight-card format

Every automated tactical insight should follow a consistent structure:

1. **Headline:** a concise observation in normal language.
1. **What happened:** the observable match behaviour.
1. **Why it matters:** the likely tactical consequence.
1. **Evidence:** match period, event count, selected metric, pitch zone or tracking sample.
1. **Confidence / coverage:** whether the conclusion is directly observed, calculated or inferred, and whether the source data is complete.
1. **Explore deeper:** opens the pitch, timeline, player or metric view behind the explanation.

Example (illustrative only):

> **The opponent found space behind the right-back.** Their left winger repeatedly received the ball in the channel after the right-back moved forward. Explore the 6 relevant sequences and the defensive shape in those moments.

Tactical glossary and explainers

Build a searchable, visual glossary for terms such as:

1. High press, mid-block and low block.
1. Build-up, progression and line-breaking pass.
1. Half-space, wide channel and zone 14.
1. Overload, underload and isolation.
1. Rest defence, counterpress and transition.
1. xG, xA, xT, EPV and PPDA.
1. Inverted fullback, false nine, double pivot and third-man run.

Each concept should include a concise definition, a pitch diagram, a real match example when licensed data is available, common misconceptions and links to related concepts.

Multiple explanation depths

For a metric such as PPDA:

1. **Simple:** “How many opposition passes a team allows before attempting a defensive action in the defined pressing area. Lower values generally indicate more aggressive pressure.”
1. **Explore:** show the team's value, opponent comparison, match periods and the events included in the calculation.
1. **Advanced:** expose the calculation definition, event filters, provider source, limitations and time-series data.

Personalised learning

1. Recommend concepts based on the matches and teams the user follows.
1. Let users save concepts and build a personal tactical glossary.
1. Link concepts to relevant match events and pitch visualisations.
1. Offer short, optional “spot the pattern” interactions based on real match sequences.
1. Track completed explainers and saved concepts; do not turn the app into a compulsory course.

AI explanation guardrails

AI can translate verified analysis into accessible language and answer follow-up questions. It must retrieve structured evidence first, cite the match and metric used, distinguish fact from interpretation, and state when data is unavailable. It must never invent tracking, video or events to fill gaps.

Signature experiences for everyday fans

These should be visible product features, not hidden under a coach-only workspace. They translate the analytical engine into questions football fans already ask.

Fan question	Product experience	Evidence / data required
Why did this team win or lose?	Match story: the 3–5 decisive tactical patterns, turning points and supporting moments	Events, score state, shots and preferably tracking
How did they dominate but fail to win?	Control vs threat: possession, territory, chance quality and dangerous transitions side by side	Events, shot data, xG where licensed
What changed after half-time?	Before/after tactical timeline with changes to formation, pressing, progression and substitutions	Events and lineups; tracking improves confidence
Why couldn't they play through the press?	Build-up explainer showing blocked routes, turnovers and the zones where possession was lost	Event coordinates, pressure events and tracking for passing lanes
Why was this player important without scoring?	Player role story: space created, defensive coverage, progression, pressing and involvement	Events plus tracking for off-ball contribution
What is this team's playing style?	Team tactical identity card showing build-up, pressing, width, directness, chance creation and defensive behaviour	Multiple matches with adequate event coverage; tracking for spatial claims
How do these two teams match up?	Tactical matchup comparison: strengths, weaknesses and likely areas of conflict	Comparable multi-match event data; tracking where available
What does this tactical term mean?	In-context visual explainer connected to an actual match sequence	Glossary plus licensed event/tracking examples

Match story layout

Each completed match should have a clear narrative flow:

1. **The short version:** one or two sentences explaining the broad tactical story without jargon.
2. **How each team tried to play:** simple formation and style descriptions.
3. **Three important patterns:** for example, one team's high press, the opponent's route around it, or a repeated overload on one flank.
4. **What changed:** substitutions, shape changes, pressing adjustments or a shift in game state.
5. **Players who shaped the game:** role-based contribution, not just goals and assists.
6. **Explore the evidence:** open the pitch, timeline, event sequence, metric definition or available video.

The app must not invent a narrative when the data is insufficient. In low-coverage matches, use a basic event-based recap and clearly state what cannot be assessed.

Team tactical identity

Create a season-level profile for every club with enough valid data. Use understandable labels alongside the underlying numbers, such as “builds patiently”, “presses high”, “attacks quickly after regaining the ball” and “creates chances from wide areas”. Avoid assigning a single simplistic style label when a team varies by opponent, score state or competition. Show the time period and sample size behind each description.

In-context tactical learning

When a user sees a term such as “half-space”, “rest defence” or “third-man run”, let them tap it to see a compact pitch diagram and a short explanation. Offer “see it in this match” where a licensed example exists. Learning should be optional and attached to the football the user is already exploring, not a separate course users must complete.

Depth preference, not separate user modes

Offer a small display preference in settings and on selected screens:

1. **Quick:** plain-language takeaway, one chart and a few key moments.
1. **Explore:** interactive pitch, player roles, phases and supporting metrics.
1. **Advanced:** detailed filters, metric definitions, tracking overlays and analyst tools when the data supports them.

Users can switch depth at any time. Do not create separate fan and coach accounts, and do not hide all professional-grade data simply because a user initially selected “Quick”.

5. Free and Pro monetization design

Launch with two plans: **Free** and **Pro**. Free must deliver genuine tactical understanding to ordinary fans, not just scores and basic statistics. Pro is for anyone who wants more history, more comparisons, richer tracking-backed detail or advanced workflows. Keep one subscription initially; do not create separate fan, coach, academy and analyst plans.

The proposed limits below are product hypotheses, not final pricing. Validate them against provider licences, data cost and user research.

Suggested plan matrix

Feature	Free	Pro
Personal account and profile	Yes	Yes
Follow clubs, competitions and players	Yes	Yes
Match centre and timeline	Yes	Yes
Basic team and player statistics	Yes	Yes, deeper history
xG and shot maps	Selected matches where licensed	Broader supported coverage and filters
Formation and basic tactical summaries	Yes	Yes, historical comparisons
Passing network	Yes	Advanced filters and comparisons
Basic pressing indicators such as PPDA	Selected supported matches	More history and phase filters
Visual tactical explainers and glossary	Full core library	Full library plus advanced examples
Selected tactical pitch overlays	Yes	More overlays and filters where tracking exists
Player role explanations	Yes	Advanced role profiles and trends
Compare teams / matches	Limited comparisons	More comparisons and longer history
Saved matches, players and insights	Basic allowance	Larger or unlimited allowance under fair use
Ask about this match (AI explainer)	Small monthly allowance, if evidence retrieval is reliable	Larger monthly allowance and longer analysis history
Advanced tracking and space analysis	A small sample or selected matches where licensed	Broader licensed coverage and deeper filters
Personal tactical model and analyst tools	Educational sample / preview	Full custom models and evaluation tools
Reports and exports	Basic shareable summary	Custom reports and eligible exports
Ads	Yes, restrained	No

Advanced features that should remain in Free

Free must offer enough depth to demonstrate that the app is genuinely about tactics:

1. Formation and tactical-shape explanation for supported matches.
1. Passing network and player involvement.
1. xG shot map where available and licensed.
1. Basic PPDA and high-turnover analysis where event definitions support it.
1. A selection of tactical pitch overlays.

1. Player role explanations and basic player comparison.
1. Visual glossary and match-linked tactical lessons.
1. Limited match comparison and saved tactical insights.
1. A small monthly AI explanation allowance after evidence retrieval is production-ready.

Do not make the free plan depend only on scores, fixtures and basic stats. Fans should be able to learn something meaningful without paying.

Pro value should come from depth and workflow

Charge for deeper historical access, advanced tracking and spatial analysis where licensed, deeper filters, more team/match/player comparisons, more AI explanations, pattern-discovery workflows, advanced reports and eligible exports. Pro should remove ads but must not be the only way to understand the tactical meaning of a match. A fan, enthusiast or coach should all be able to understand the Free/Pro difference without learning professional coaching terminology.

Usage entitlements

Do not hardcode plan names throughout the app. Implement a server-side entitlement system:

```
Entitlement
- user_id
- feature_key
- limit_value
- usage_period
- valid_from
- valid_until
- source_plan
- status
```

Examples of `feature_key`:

1. `following.max_saved_items` (only if a limit is genuinely needed)
1. `analysis.matches_per_month`
1. `ai.credits_per_month`
1. `reports.exports_per_month`
1. `tracking.full_access`
1. `video.storage_gb`
1. `exports.bulk_enabled`

The backend must enforce entitlements. Hiding a button in Android is not sufficient access control. Do not arbitrarily limit basic following of clubs or access to public fixture information just to force conversion.

6. Ads integration and ad experience

Recommended ad network

Use Google AdMob for the first release. Start with a small number of placements and add mediation only after the app has enough traffic to justify it.

Suitable ad placements

1. A native ad card in Home or Explore after useful content, not between every insight.
1. A small banner on low-intent browsing screens only where it does not compete with football content.
1. Avoid rewarded ads for core tactical insights; do not make fans watch an ad to understand a match.

Screens that should remain ad-free

1. Sign-up and login.
1. Payment and subscription management.
1. Live match or match-review workflow.
1. Tactical pitch and video analysis canvas.

1. Report editing/export flow.
1. Privacy, security and account deletion.

Ad implementation requirements

1. Integrate AdMob through a dedicated `AdsRepository` / adapter so ad networks can be changed later.
1. Use Google's User Messaging Platform (UMP) for consent and privacy choices.
1. Refresh consent status at app launch and show required consent flows before loading personalized ads.
1. Provide a visible privacy-options entry point when required.
1. Keep ad identifiers and analytics permissions separate from core account authentication.
1. Do not send player-performance data, tactical notes or private team information to ad networks.
1. Disable ads automatically for paid entitlements.
1. Use test ad units in development and staging.
1. Monitor ad load failures and fill rate; do not let ad failures block the main app.

Official guidance: https://developers.google.com/admob/android/privacy

7. Authentication and account security

Android sign-in methods

Recommended order:

1. Sign in with Google through Android Credential Manager.
2. Passkeys through Credential Manager.
3. Email/password or email magic-link fallback.
4. Optional phone OTP for recovery or phone verification, not the only authentication path.

Avoid requiring a phone number before the user can see the product. Phone OTP introduces SMS cost, delivery failure and SIM-swap risk. Use a trusted authentication provider or a well-reviewed implementation; do not implement password hashing, recovery or OTP flows casually.

Official Android reference: https://developer.android.com/identity/credential-manager

Backend session design

1. Backend verifies the Google/passkey/email identity token.
1. Issue a short-lived access token (e.g. 5–15 minutes).
1. Use rotating refresh tokens with reuse detection and revocation.
1. Store refresh-token hashes and device/session metadata server-side.
1. Allow users to review and revoke active sessions.
1. Store sensitive client tokens using Android Keystore-backed secure storage.
1. Never embed provider API keys, payment secrets or admin credentials in the APK.

Account lifecycle

1. Account linking for Google/email/passkey identities.
1. Email verification and recovery flow.
1. Optional phone verification.
1. Device/session management.
1. Account export and deletion workflow.
1. Follow/unfollow real-world clubs, national teams, competitions and players.

1. Optional custom-team create, rename, archive and delete workflow only if the later local/academy feature is introduced.
1. Personal data export and retention controls.

Abuse protection

1. IP and account-based rate limits.
1. OTP resend limits, cooldowns and fraud checks.
1. Login throttling and suspicious session detection.
1. Play Integrity signals on high-risk requests as an anti-abuse signal, not as a replacement for authentication.
1. CAPTCHA or risk-based challenge where appropriate.
1. WAF/API gateway rules and request-size limits.

Private-data authorization

Use ownership checks, not a general RBAC framework:

1. Derive `user_id` from the verified access token.
1. For user-private records, include `owner_user_id = authenticated_user_id`. Do not apply ownership checks to shared real-world club records; those are canonical read-only football entities. Apply ownership checks only to optional custom teams and user-created resources.
1. Never trust a `user_id`, `owner_user_id` or subscription status supplied by the Android client.
1. Return `404` for resources the user does not own where this avoids leaking resource existence.
1. Test cross-account access for every private endpoint.

Play Integrity reference: https://developer.android.com/google/play/integrity/standard

8. Payments and subscriptions

Default Android payment system

Use Google Play Billing for digital subscriptions and paid app functionality distributed through Google Play, unless a specific policy exception or enrolled program applies.

Use the latest stable Google Play Billing Library version supported by the app's target SDK and Play requirements at implementation time. Pin the dependency, review release notes before each major release and keep a regular security/dependency upgrade process.

References:

1. Billing integration: https://developer.android.com/google/play/billing/integrate
1. Release notes: https://developer.android.com/google/play/billing/release-notes
1. Payments policy: https://support.google.com/googleplay/android-developer/answer/9858738

Subscription products

Start with a manageable catalogue:

1. `free` — ad-supported.
1. `pro_monthly` — individual Pro monthly subscription.
1. `pro_annual` — individual Pro annual subscription.

Do not launch too many confusing tiers. Start with Free and Pro monthly/annual products that make sense to any user, not a coach-only subscription. Pro should sell more depth, history, analysis capacity and workflow tools. Consider a trial only after provider and AI usage costs are understood. A future multi-user club/academy plan is out of scope until collaboration is proven necessary. Show price, billing period, renewal behaviour, trial conversion date and cancellation instructions clearly.

Purchase lifecycle

```
Android app
  -> Query product details from Play
  -> Launch Google Play purchase UI
  -> Receive purchase token
  -> Send token + product ID + authenticated user to backend
  -> Backend verifies purchase with Google Play Developer API
  -> Backend records purchase and updates entitlements
  -> Backend acknowledges purchase as required
  -> Android refreshes account entitlements

Google Play RTDN
  -> Pub/Sub notification
  -> Backend validates notification
  -> Backend queries Google Play Developer API for complete state
  -> Update subscription state and entitlements idempotently
```

Important implementation requirements:

1. Never trust a client-only purchase success callback.
1. Verify purchase tokens on the backend.
1. Make purchase processing idempotent.
1. Handle pending, active, grace period, on-hold, paused, cancelled, refunded and expired states.
1. A user who cancels should retain access until the paid-through date if the subscription state permits it.
1. Implement restore purchases and cross-device entitlement refresh.
1. Keep billing history and entitlement history auditable.
1. Reconcile subscriptions periodically in case a notification is delayed or missed.

Google Play RTDN documentation explicitly requires the backend to call the Google Play Developer API after a notification to retrieve complete subscription state: https://developer.android.com/google/play/billing/rtdn-reference

India-specific billing option

Google documents an alternative billing program for eligible developers serving users in India, with additional onboarding, reporting, safety and PCI DSS requirements. Do not assume a normal payment gateway can simply replace Play Billing. Start with Play Billing; consider the India alternative-billing program only after checking current eligibility and economics with legal/finance.

Reference: https://support.google.com/googleplay/android-developer/answer/13306652

9. Football data provider strategy

There is no single provider that guarantees all event, tracking, video and physical data for every competition. Separate these capabilities explicitly.

Recommended provider roles

Provider	Proposed role	Useful data	Main limitation / decision
Sportmonks Football API	Initial fixtures/events/stats provider	Fixtures, teams, players, lineups, events, statistics, xG/pressure add-ons and ball coordinates depending on plan	Good starting integration; ball coordinates are not full continuous player tracking
StatsBomb Open Data	Development and analytics research	Historical events, lineups and selected 360 freeze-frames	Selected matches/competitions; open-data terms are not a blanket commercial redistribution licence
SkillCorner Open Data	Tracking engine development	10 sample 2024/25 A-League games, tracking, dynamic events, phase data and season aggregates	Sample data only; tracking includes known identity/noise limitations
SkillCorner commercial	Potential tracking vendor	Broadcast player tracking, physical/dynamic metrics and APIs	Obtain coverage, delivery, derived-metric and commercial-use quote

Provider	Proposed role	Useful data	Main limitation / decision
Opta Vision	Enterprise event + tracking alternative	Continuous XY positions for all 22 players synchronized to events, phases and off-ball analysis	Premium enterprise relationship; request demo and licence
Sportradar	Alternative broad coverage/live feed	Competitions, live events, lineups, stats and ball spotting; extended stats package	Confirm exact coverage and data rights; ball spotting is not 22-player tracking
API-Football	Optional prototype/fallback feed	Fixtures, events, lineups and player/team data	Do not treat as a tactical-tracking source

Official references:

1. Sportmonks: https://www.sportmonks.com/football-api/ and https://docs.sportmonks.com/
1. StatsBomb Open Data: https://github.com/hudl/open-data
1. SkillCorner Open Data: https://github.com/SkillCorner/opendata and https://skillcorner.github.io/opendata/
1. SkillCorner API: https://skillcorner.com/api/docs/
1. Opta Vision: https://www.statsperform.com/products/opta-vision/
1. Sportradar Soccer API: https://developer.sportradar.com/soccer/reference/soccer-overview
1. API-Football: https://www.api-football.com/

Important licence gate before production

For each data vendor, obtain written answers to all of these:

1. Can the feed be used in a public commercial Android application?
2. Can licensed data be displayed to individual coaches and end users, and under what redistribution restrictions?
3. Can raw events, tracking and derived metrics be stored? For how long?
4. Can the product generate and sell derived tactical metrics?
5. Can data be used to train internal ML models?
6. Can generated reports and clips be exported or shared?
7. Are there user, club, territory, competition or device restrictions?
8. What is the permitted attribution and logo requirement?
9. Is live redistribution permitted, and what latency/refresh limits apply?
10. What is the provider's policy on caching, deletion, termination and audit?

Do not ship StatsBomb Open Data or SkillCorner sample files as a commercial customer-facing dataset until the exact dataset terms and any separate data rights have been reviewed. Use them to build, test and validate the analytics pipeline, and use properly licensed feeds for production customer data.

Recommended staged procurement

Stage 1 — Development: StatsBomb Open Data + SkillCorner Open Data + Sportmonks free/developer access.

Stage 2 — Initial public production: select one commercial event/fixture API after checking target league coverage, commercial app rights, historical depth, rate limits and actual event fields. Sportmonks is a practical first candidate to evaluate.

Stage 3 — Advanced tactical production: contract one full player-tracking feed (SkillCorner or Opta Vision) for a limited competition set. Do not buy multiple tracking feeds until a concrete product requirement justifies it.

Stage 4 — Enterprise expansion: consider Sportradar/Opta or additional feeds where customers require competition coverage, latency, data guarantees or redundancy.

Data capability matrix

Feature	Event data	Event + 360 freeze-frame	Continuous tracking	Video
Fixtures, results, lineups	Yes	Yes	Yes	Optional

Feature	Event data	Event + 360 freeze-frame	Continuous tracking	Video
Shot maps and basic xG	Yes	Yes	Yes	Optional
Passing network	Yes	Yes	Yes	Optional
Progressive passes/carries	Usually	Yes	Yes	Optional
Local passing options at event	No	Partial	Yes	Optional
Team width/length through time	No	Snapshot only	Yes	Helpful
Defensive line and compactness	No	Limited snapshots	Yes	Helpful
Off-ball runs	No	Limited snapshot inference	Yes	Helpful
Space occupation / control	No	Partial	Yes	Helpful
Physical speed/acceleration	No	No	Yes, subject to feed quality	Optional
Tactical video clips	No	No	Timestamp sync required	Yes

Treat data availability as a runtime capability, not an assumption. For example, do not show “defensive compactness” if a match only contains event data.

10. Canonical football data model

Keep shared football records separate from user-private data. Real-world clubs and competitions are global provider entities. A user follows them; the user does not own or duplicate them.

Shared football entities

1. `Competition`
1. `Season`
1. `Team` (real-world club or national team, provider-mapped)
1. `Player`
1. `PlayerTeamAssignment`
1. `Match`
1. `MatchParticipant`
1. `Lineup`
1. `Event`
1. `TrackingAsset`
1. `TrackingFrame` (stored in analytical/object storage, not one Postgres row per coordinate long term)
1. `PhaseOfPlay`
1. `TacticalMetric`
1. `MetricDefinition`

User-private entities

1. `User`
1. `UserIdentity`
1. `UserSession`
1. `FollowedTeam`
1. `FollowedCompetition`
1. `FollowedPlayer`

1. `SavedMatch`
1. `SavedInsight`
1. `UserCollection`
1. `UserNote`
1. `LearningProgress`
1. `GameModel` (optional advanced workspace, owned by a user)
1. `GameModelPrinciple`
1. `Report` (owned by a user)
1. `VideoAsset` / `VideoClip` (only where rights permit)
1. `Subscription`
1. `Entitlement`
1. `SecurityEvent`
1. `CustomTeam` (optional later feature for local/academy teams, with `owner_user_id`)

Relationship model

```
Team (shared canonical football entity)
  ■■■ Competition / season participation
  ■■■ Squad and players
  ■■■ Matches

User
  ■■■ FollowedTeam -> Team
  ■■■ FollowedCompetition -> Competition
  ■■■ FollowedPlayer -> Player
  ■■■ SavedMatch -> Match
  ■■■ SavedInsight -> TacticalMetric / insight reference
  ■■■ UserNote -> Match / Team / Player / Insight
  ■■■ Optional CustomTeam (private, owner_user_id)
```

Normalization requirements

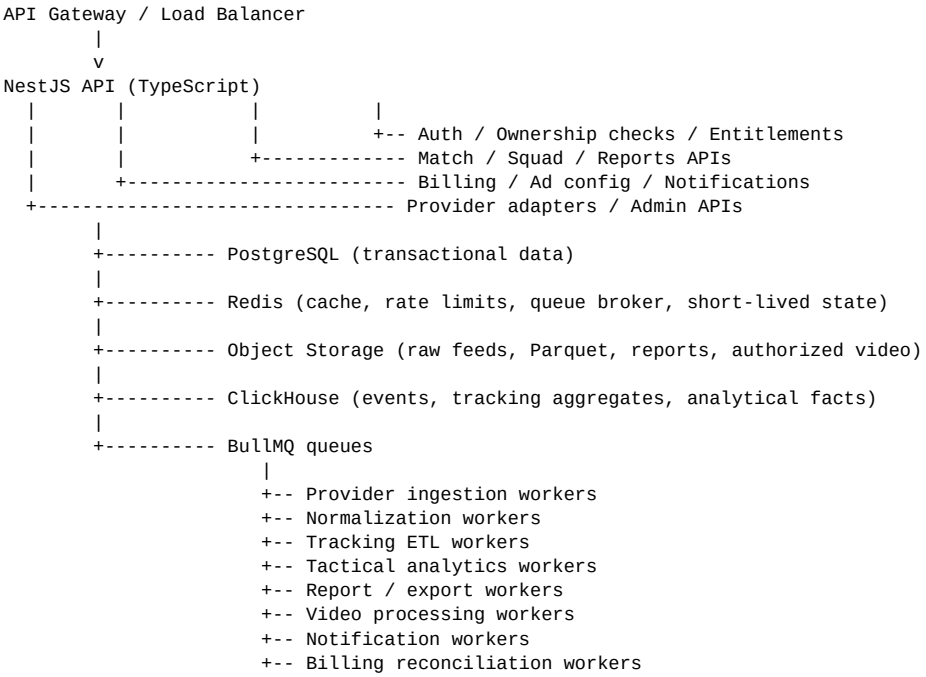
1. Normalize pitch orientation to one canonical attacking direction.
1. Store pitch dimensions and units.
1. Keep raw provider coordinates and normalized coordinates where necessary.
1. Normalize match clock, period, stoppage time and video timecode separately.
1. Keep event taxonomy mappings versioned.
1. Maintain provider player/team ID mapping tables.
1. Preserve event and tracking confidence/quality flags.
1. Handle missing, extrapolated and occluded tracking points.
1. Store source metadata, licence identifier and retrieval timestamp.
1. Never assume the same metric definition across vendors. Store a metric-definition version.
1. Make feature availability queryable per match so the UI can explain whether event, freeze-frame, tracking and video data exist.

11. Backend architecture

Recommended initial architecture: modular monolith + worker fleet

Use a single backend repository with independently deployable API and worker processes. This is easier to maintain than starting with many microservices, but provides queue isolation and horizontal scaling.

```
Android App (Kotlin / Compose)
        |
        | HTTPS REST / WebSocket
        v
```



Responsibilities by component

NestJS API

1. Authentication and sessions.
1. User-owned team access checks.
1. Match, squad, player and report endpoints.
1. Provider adapter orchestration.
1. Billing and entitlement endpoints.
1. WebSocket/SSE gateway for job status and live updates.
1. Admin and support tools.

PostgreSQL

1. Users, identities, user-owned teams, squads and players.
1. Match metadata and provider mapping.
1. Game models, notes, reports and video metadata owned by the user.
1. Purchases, subscriptions, per-user entitlements and security events.
1. Data ingestion job metadata and idempotency records.

Redis

1. BullMQ queue backend.
1. Short-lived cache for frequently requested match summaries and dashboard payloads.
1. API rate limits and abuse counters.
1. Distributed locks for scheduled provider syncs.
1. Ephemeral WebSocket fan-out/presence where needed.
1. Avoid storing durable billing state or irreplaceable user data only in Redis.

ClickHouse

1. Large-scale event facts and analytical aggregates.
1. Tracking-derived player/team spatial metrics.
1. Season/player/role trend aggregations.

1. Fast filtering across matches, events, teams and time ranges.

Object storage

1. Raw provider payloads when licence permits.
1. JSONL/Parquet tracking files.
1. Generated PDFs, thumbnails and report artifacts.
1. Authorized user-uploaded video and derived clips.
1. Use lifecycle/retention policies and user-scoped object keys.

Python/Rust analytics services

1. Python + Polars for feature engineering, ML experiments and data science workflows.
1. Rust for CPU/memory-intensive tracking transforms or streaming parsers if profiling proves it is beneficial.
1. Keep analytics workers callable through queue jobs or internal APIs; don't expose them directly to the public internet.

Why not use Redis as the primary analytics database?

Redis is excellent for queues, caching, short-lived coordination and rate limiting. It is not the long-term source of truth for match events or tracking history. Store transactional data in PostgreSQL, analytical facts in ClickHouse, and large files in object storage.

12. Redis and worker queue design

Use BullMQ initially because the recommended NestJS backend is TypeScript and BullMQ integrates naturally with Redis. Separate workers by workload and resource profile.

Queues

Queue	Example jobs	Worker profile
`provider-sync`	Refresh fixtures, squads, competitions, player profiles	I/O-bound
`match-ingestion`	Fetch events, lineups, match statistics	I/O-bound
`normalization`	Map provider payloads into canonical records	CPU-light / moderate
`tracking-etl`	Parse JSONL, normalize frames, calculate movement features	High memory / CPU
`tactical-analytics`	Shape, pressing, transition, possession metrics	CPU-bound
`pattern-discovery`	Find repeated tactical sequences and cluster patterns	CPU/GPU depending on model
`report-generation`	Build PDF, charts, tables and report snapshots	CPU/moderate
`video-processing`	Authorized transcodes, thumbnails, clip extraction	High CPU/I/O
`notifications`	FCM push and email delivery	I/O-bound
`billing-reconciliation`	Verify purchases and reconcile subscription state	I/O-bound / high correctness
`data-retention`	Delete/expire source data according to licence and user policy	I/O-bound

Queue job envelope

```
{
  "job_id": "stable-idempotency-key",
  "job_type": "ingest_match_events",
  "requested_by_user_id": "authenticated-user-id-or-null-for-system-sync",
  "scope": "global-provider-sync-or-user-request",
  "provider": "provider-name",
  "provider_match_id": "source-id",
  "schema_version": 1,
  "requested_at": "2026-11-11T00:00:00Z",
  "trace_id": "trace-id"
}
```

Reliability rules

1. Every job must be idempotent; retries must not duplicate rows, payments or reports.
1. Use deterministic job IDs for provider syncs and per-match processing.
1. Configure exponential backoff with jitter for transient failures.
1. Use a dead-letter/failed-job workflow and admin replay tools.
1. Keep provider rate limits in a shared limiter, not per worker process only.
1. Limit concurrency separately for event ingestion, tracking processing and video work.
1. Use bounded payloads; pass object-storage references instead of huge tracking arrays through Redis.
1. Persist job state and progress in PostgreSQL where it is user-visible or needed for recovery.
1. Use checkpoints for large tracking files and resumable multipart uploads.
1. Use queue priorities for billing events, user-triggered jobs and routine backfills.
1. Apply retention and cleanup rules to completed BullMQ jobs.
1. Monitor queue depth, oldest-job age, failure rate, retry count and worker saturation.

Live updates

For live match features, ingest the provider's push/stream feed where contractually available. Normalize incoming events, store them, then publish lightweight updates through Redis pub/sub or a dedicated stream to WebSocket clients. Do not poll an expensive API for every connected user.

13. API surface (illustrative)

Use versioned REST endpoints with OpenAPI documentation.

Authentication and profile

1. `POST /v1/auth/exchange`
1. `POST /v1/auth/refresh`
1. `POST /v1/auth/logout`
1. `GET /v1/me`
1. `GET /v1/me/sessions`
1. `DELETE /v1/me/sessions/:sessionId`
1. `DELETE /v1/me`

Football discovery and following

1. `GET /v1/competitions`
1. `GET /v1/competitions/:competitionId`
1. `GET /v1/teams?competitionId=...&seasonId=...` (shared provider-backed club catalogue)
1. `GET /v1/teams/:teamId`

```
GET /v1/teams/:teamId/players
```

```
GET /v1/teams/:teamId/coverage
```
(available seasons and data types: fixtures, events, tracking, video)

```
GET /v1/players/:playerId
GET /v1/me/following
PUT /v1/me/following/teams/:teamId
DELETE /v1/me/following/teams/:teamId
PUT /v1/me/following/competitions/:competitionId
DELETE /v1/me/following/competitions/:competitionId
PUT /v1/me/following/players/:playerId
DELETE /v1/me/following/players/:playerId
```

Personal saved items

```
GET /v1/me/saved-matches
PUT /v1/me/saved-matches/:matchId
DELETE /v1/me/saved-matches/:matchId
GET /v1/me/saved-insights
POST /v1/me/notes
GET /v1/me/collections
```

All private saved-item and note endpoints verify that the authenticated user owns the resource. Shared football records remain provider-mapped canonical entities. Optional custom-team endpoints can be added later if local/academy users validate that need.

Matches and analytics

```
GET /v1/matches
GET /v1/matches/:matchId
GET /v1/matches/:matchId/events
GET /v1/matches/:matchId/metrics
GET /v1/matches/:matchId/tactical-shape
GET /v1/matches/:matchId/pressing
GET /v1/matches/:matchId/transitions
GET /v1/matches/:matchId/players/:playerId
POST /v1/matches/:matchId/analysis-jobs
GET /v1/jobs/:jobId
```

Game model and reports

```
POST /v1/game-models
PATCH /v1/game-models/:id
POST /v1/game-models/:id/evaluate
POST /v1/reports
GET /v1/reports/:id
GET /v1/reports/:id/export
POST /v1/video-clips
```

Billing and entitlements

```
GET /v1/billing/products
POST /v1/billing/google-play/verify
GET /v1/billing/subscription
GET /v1/entitlements
POST /v1/billing/restore
POST /v1/webhooks/google-play-rtdn
```

All list endpoints must support pagination, filtering and bounded page sizes. All private user-resource endpoints must apply a centralized ownership check using the authenticated user ID. Shared real-world club, league, player and match endpoints use canonical IDs and provider coverage rules, not per-user ownership.

14. Android app architecture

Recommended stack

- Kotlin.
- Jetpack Compose and Material 3.
- Navigation 3 or the supported Navigation component at implementation time.
- ViewModel + StateFlow.
- Kotlin Coroutines and Flow.
- Hilt for dependency injection.
- Retrofit/OkHttp or Ktor client for networking.
- Room for local structured cache/offline data.
- DataStore for non-sensitive preferences.
- Keystore-backed storage for refresh credentials.
- WorkManager for deferred syncs and uploads.
- Firebase Cloud Messaging for push notifications.
- Play Billing Library for subscriptions.
- AdMob + UMP for ads and consent.

Suggested module layout

```
android/
■■■ app/
■■■ core/
■   ■■■ common/
■   ■■■ designsystem/
■   ■■■ network/
■   ■■■ database/
■   ■■■ auth/
■   ■■■ billing/
■   ■■■ ads/
■   ■■■ analytics/
■■■ feature/
■   ■■■ onboarding/
■   ■■■ home/
■   ■■■ explore/
■   ■■■ following/
■   ■■■ matches/
■   ■■■ matchcentre/
■   ■■■ tacticalpitch/
■   ■■■ clubprofile/
■   ■■■ playerprofile/
■   ■■■ tacticallearning/
■   ■■■ advancedanalysis/
■   ■■■ gamemodel/       # optional advanced workspace
■   ■■■ reports/
■   ■■■ subscription/
■   ■■■ settings/
■■■ build-logic/
```

UI state and offline behaviour

1. ViewModels expose immutable screen state.
1. UI sends user actions to ViewModels; composables do not call data sources directly.
1. Cache recently viewed match summaries and squad data.
1. Show stale-data timestamps clearly.
1. Queue notes and low-risk edits for sync when offline, with conflict handling.
1. Do not cache licensed raw tracking/video locally unless the contract permits it.
1. Use pagination and lazy lists for match history and event timelines.
1. Render tactical pitch layers efficiently; avoid recomputing hundreds of objects during every Compose recomposition.
1. For large frame sequences, use precomputed/downsampled representations and a dedicated rendering layer.
1. Test on lower-memory Android devices as well as flagship phones.

15. Analytics and ML implementation strategy

Build deterministic metrics first, then add machine learning where it creates real value.

Phase 1: deterministic analytics

1. Coordinate normalization and pitch zones.
1. Event sequences and possession chains.
1. Passing networks and progression.
1. Basic xT grid and possession-value baselines.
1. Team shape snapshots from tracking.
1. Width, length, compactness and line distances.
1. Speed/acceleration smoothing with quality flags.
1. Pressing and transition windows.

Phase 2: tactical classification

1. Possession-phase classification.
1. Build-up sequence classification.
1. Formation and role detection.
1. Pressing trigger classification.
1. Off-ball run classification.
1. Set-piece routine classification.
1. Repeated pattern clustering.

Phase 3: learned models

1. Role-specific player archetypes.
1. Pass option/value models.
1. Opponent style similarity.
1. Tactical pattern discovery.
1. Game-model adherence scoring.
1. Recommendation ranking based on evidence and coach feedback.

Model quality rules

1. Define ground truth and evaluation splits before training.
1. Separate train/test by match or season, not random frames from the same match.
1. Avoid leakage from event labels or post-match outcome fields.
1. Validate across different competitions and tracking providers.
1. Report confidence and sample size.
1. Monitor drift and provider schema changes.
1. Keep human review for pattern labels and consequential recommendations.
1. Version models, features and metric definitions.

AI analyst guardrails

1. Query structured analytics first; do not ask the LLM to calculate raw metrics from prose.
1. Ground every numerical claim in a database result.
1. Attach match, time range, data source and methodology to evidence.
1. Do not output private data belonging to another user.
1. Refuse unsupported questions rather than inventing tracking/video facts.
1. Keep model output auditable with request ID, model version, retrieved evidence IDs and user feedback.

16. Security, privacy and production readiness

Backend security

1. TLS everywhere; HSTS for web-facing endpoints.
1. Secrets in a managed secrets service, never in source control or Android assets.
1. Strong password hashing if passwords are managed directly (Argon2id or an established identity provider).
1. Access-token rotation and refresh-token reuse detection.
1. User-level ownership enforced in the repository/service layer; add PostgreSQL row-level security only if it materially simplifies and strengthens isolation.
1. Ownership checks on every private read, write, export and video action.

- Server-side validation with strict schema and size limits.
- SQL parameterization and ORM/query safety.
- API rate limits by IP, authenticated user and endpoint class.
- Signed webhooks and replay protection.
- Idempotency keys for purchases, uploads and job creation.
- Encryption at rest for databases and object storage; additional field-level encryption for particularly sensitive data.
- Security logs for account sessions, purchase verification, exports and account changes; avoid building a broad enterprise audit system in v1.
- Regular backup and restore drills.
- Dependency/container scanning, SAST, secret scanning and penetration tests.

Android security

- Use Credential Manager for modern authentication.
- Keystore-backed storage for sensitive credentials.
- Network security configuration; no cleartext traffic in production.
- Do not store provider API keys or billing secrets in the APK.
- Obfuscate/release-build hardening and remove debug endpoints.
- Play Integrity as an anti-abuse signal for sensitive actions.
- Protect deep links and exported components.
- Avoid logging tokens, personal data or private tactical notes.
- Use screenshot/screen-recording protection only for genuinely sensitive screens and after evaluating user impact; it is not a substitute for server authorization.

Privacy and data governance

- Publish clear privacy policy and in-app privacy controls.
- Collect only data necessary for account, team management and analytics.
- Define retention periods for raw feeds, generated reports and user-uploaded footage.
- Provide account export and deletion workflows.
- Obtain appropriate permissions for player data, especially youth academy records and video.
- Do not send tactical data or player-level performance to advertising systems.
- Review applicable India DPDP requirements and any other regions where the app is offered with qualified legal counsel.
- Perform a privacy and data-protection review before launch.

Use OWASP MASVS/MASTG as the mobile app security verification baseline:
https://owasp.org/projects/mobile-application-security

17. Deployment and operations

Suggested AWS production layout

```
Route 53 / DNS
   |
CloudFront (static assets, public downloads if applicable)
   |
Application Load Balancer
   |
ECS/Fargate services
   ■■■ NestJS API service (min 2 tasks for production HA)
   ■■■ WebSocket/API realtime service (if separated)
   ■■■ Ingestion workers
   ■■■ Analytics workers
   ■■■ Report/video workers
   ■■■ Scheduled jobs

Managed services
   ■■■ RDS PostgreSQL
   ■■■ ElastiCache Redis
   ■■■ S3 object storage
   ■■■ ClickHouse Cloud or managed analytical database
   ■■■ Secrets Manager
   ■■■ CloudWatch / OpenTelemetry collector
   ■■■ FCM / Google Play integrations
```

Start with one region close to the primary customer base, and select data residency and backup locations based on customer contracts and applicable law. Keep deployment cloud-agnostic where practical through containers and infrastructure-as-code.

Environments

1. Local development with Docker Compose.
1. Shared development environment.
1. Staging with Play Billing test products and AdMob test units.
1. Production with separate secrets, buckets, databases and provider keys.

Observability

1. Structured JSON logs with correlation/trace IDs.
1. OpenTelemetry distributed traces.
1. API latency, error rate and throughput dashboards.
1. Queue depth, oldest job age, worker concurrency and retry dashboards.
1. Provider quota usage, latency, errors and data freshness.
1. ClickHouse query latency and storage growth.
1. Billing verification failures and entitlement mismatch alerts.
1. Crashlytics for Android crashes and ANRs.
1. Synthetic tests for sign-in, match load, report generation and subscription restore.

Service targets to validate

Area	Initial target
Core API availability	99.9% monthly after production stabilization
Normal API p95	Under 500 ms for cached/light queries
Match summary p95	Under 1.5 s for common queries
Dashboard first useful render	Under 2.5 s on typical supported network/device
User-triggered job status	Visible within 2 s after enqueue under normal load
Billing entitlement update	Near-real-time after verified purchase/notification
Backup restore test	Successful quarterly restore drill

These are engineering targets, not guarantees. Benchmark with real workloads and adjust as traffic and vendor feeds grow.

18. Testing strategy

Android

1. Unit tests for ViewModels, repositories, billing state handling and entitlement rendering.
1. Compose UI tests for sign-in, onboarding, match review, paywall and settings.
1. Navigation regression tests.
1. Accessibility checks and TalkBack testing.
1. Macrobenchmark startup, scrolling and tactical-pitch rendering.
1. Low-end device, low-memory and weak-network testing.
1. Offline mode and process-death recovery tests.

Backend

1. Contract tests for every provider adapter.
1. Schema validation against versioned provider payload samples.
1. Cross-account ownership and authorization tests.
1. Purchase verification and RTDN replay/idempotency tests.
1. Queue retry, dead-letter and worker restart tests.
1. Data-ingestion replay tests.
1. Load tests for match list, match centre, analytics endpoints and WebSocket fan-out.
1. Backup/restore and disaster-recovery tests.

Analytics

1. Hand-checked match samples for key metrics.
1. Tracking quality and missing-frame checks.
1. Cross-provider coordinate normalization tests.
1. Unit tests for pitch orientation and match-clock alignment.
1. Model evaluation split by match/season.
1. Explainability tests for generated insights.
1. Human analyst review before enabling automatic tactical recommendations broadly.

19. Development roadmap

A practical roadmap is milestone-based. The exact duration depends on team size, provider onboarding and data licensing; do not commit to live tracking or AI features before the data feed is secured.

Milestone 0 — Fan-first product and data validation

Deliverables

1. Product requirements led by everyday football fans, then tactical enthusiasts and professional users.
1. User journeys and low-fidelity screen flows for Home, Explore, Match Story, Club, Player and Tactical Learning.
1. Provider comparison and written licence checklist.
1. Download and parse StatsBomb Open Data and SkillCorner Open Data.
1. Test Sportmonks developer access and inspect actual event/stat fields.

1. Build a small canonical event schema and tracking parser.
1. Validate one complete end-to-end sample match.

Exit criteria

1. Know which features are possible with event data only.
1. Know which features require full tracking.
1. A provider and target competition shortlist exists.
1. Commercial data rights are understood before production commitments.

Milestone 1 — Android foundation, login and fan personalisation

Deliverables

1. Kotlin/Compose app shell and design system.
1. Credential Manager sign-in and backend session exchange.
1. User profile and simple following lists for clubs, competitions and players.
1. Onboarding built around football interests, not organization or squad creation.
1. Private saved matches, notes, bookmarks and collections.
1. Ownership checks on all private user data.
1. Sample/demo match story and interactive tactical explainer for new users.
1. Room cache and account settings.

Exit criteria

1. User can sign in, follow clubs from supported competitions, explore a match and save an insight.
1. A user cannot access another user's private notes, saved lists, reports or uploads.
1. No organization, membership, staff-seat, invitation or custom permission system is required for v1.

Milestone 2 — Data ingestion, discovery and match centre

Deliverables

1. Provider adapter interface.
1. Sportmonks integration or selected contracted feed.
1. Scheduled fixture/team/player sync.
1. Match event ingestion and canonical normalization.
1. PostgreSQL + object storage + Redis + BullMQ.
1. Match list, match centre, timeline, lineups, basic stats, shot map and plain-language match story.

Exit criteria

1. Provider updates are idempotent.
1. Quota-aware ingestion and retry handling work.
1. The Android app can display a match from normalized backend data.

Milestone 3 — Fan-readable tactical analytics foundation

Deliverables

1. Pitch coordinate normalization.
1. Passing networks and progressive actions.
1. Build-up and possession sequence analysis.
1. xT / possession value baseline.

1. PPDA and high-turnover metrics with definitions.
1. Basic player role profiles.
1. Match comparison, team-style summaries and evidence-linked plain-language insights.

Exit criteria

1. Every displayed metric has a documented definition, source and data-quality label.
1. Analysts validate sample outputs against source matches.

Milestone 4 — Tracking and advanced team-shape analytics

Prerequisite: signed commercial tracking agreement or approved user-provided tracking data.

Deliverables

1. Tracking ingestion and quality validation.
1. ClickHouse event/tracking storage.
1. Dynamic formations and phases of play.
1. Compactness, width, length and line-distance metrics.
1. Off-ball runs and space occupation.
1. Pressing and transition analysis.
1. Tactical pitch visualization.

Exit criteria

1. Frame/time synchronization passes validation.
1. Missing/low-confidence tracking is surfaced rather than hidden.
1. Tracking analytics scale in worker tests.

Milestone 5 — Personal reports and video

Deliverables

1. Report templates and PDF generation.
1. Personal notes and annotations.
1. Private video playlists and saved match moments where rights permit. Defer shared team workspaces and multi-user collaboration.
1. Authorized video upload/integration.
1. Clip extraction and pitch/video synchronization.
1. FCM notifications for completed analysis jobs and report generation.

Exit criteria

1. Reports and clips are owner-isolated and ownership-checked.
1. Video storage and exports comply with source rights.

Milestone 6 — Freemium, ads and billing

Deliverables

1. Server-side entitlement engine.
1. Free/Pro plans and per-user usage limits validated against real fan usage and provider cost.
1. Google Play Billing integration.
1. Server-side purchase verification and RTDN.
1. Subscription restore and management UX.
1. AdMob and UMP consent.

1. Ad-free paid entitlement and placement policy.

Exit criteria

1. No paid feature can be unlocked by modifying client state.
1. Purchase/refund/cancel/grace-period test cases pass.
1. Ads never interrupt match stories, tactical exploration, video review or payment flows.

Milestone 7 — Tactical intelligence and AI match explainer

Deliverables

1. Optional personal tactical model for advanced users.
1. Pattern discovery and analyst review workflow.
1. Evidence-grounded “Ask about this match” experience for fans and advanced users.
1. Opponent reports and automatic tactical summaries.
1. What-if scenario workspace.
1. Model quality and usage monitoring.

Exit criteria

1. AI answers cite structured evidence and clearly communicate uncertainty.
1. No unsupported tracking/video statements are generated.
1. Users can rate, correct or dismiss discovered patterns; professional users can add detailed annotations.

Milestone 8 — Production launch and scale

Deliverables

1. Security audit and penetration test.
1. Load and reliability tests.
1. Provider contract and data rights sign-off.
1. Privacy policy, terms, deletion and retention processes.
1. Play Console release tracks and staged rollout.
1. Operational dashboards, alerts and incident runbooks.
1. Customer support and billing dispute procedures.

Exit criteria

1. Rollback tested.
1. Backup restore tested.
1. Critical security findings closed.
1. Data provider coverage and freshness accurately represented to users.

20. MVP scope: what to ship first

Do not try to ship every tracking and coaching feature in the first release. Ship a fan-first tactical exploration product that is genuinely useful, with enough depth to attract tactical enthusiasts and a clear path to professional tools.

First public MVP

1. Secure sign-in and a personalised football profile.
2. Follow any supported clubs, competitions and players; do not require a user-created team.
3. Search and browse competition, club, player and match pages.
4. Fixtures, match centre, match timeline, lineups and core statistics.

5. A readable post-match story: how both teams approached the game, three important patterns, key turning points and what changed after half-time.
6. Basic event-based tactical explanations: formation, passing network, progression, shot map and xG where licensed.
7. A visual tactical glossary that explains common tactical concepts in plain language.
8. A Quick / Explore / Advanced view on selected tactical metrics.
9. Evidence-backed match insights with links to events and pitch zones.
10. Save matches, players and tactical insights.
11. A useful free tier and a simple Pro upgrade; ads only in low-interruption discovery areas.
12. Product analytics measuring whether users understand and revisit tactical insights.

Advanced features to add after validating the core experience

1. Continuous tracking-backed team shape, compactness and space occupation.
1. Off-ball runs and role-aware player analysis.
1. Advanced pressing, counterpress and transition analysis.
1. Automated pattern discovery and opponent reports.
1. AI match explainer with evidence retrieval and follow-up questions.
1. Optional professional analysis workspace, personal tactical models and advanced reports.
1. Video synchronization and clips where the required rights are available.

Do not put these in the first public MVP unless data and rights are already secured

1. Full continuous tracking on every match.
1. Reliable physical-load monitoring from broadcast-only data.
1. Live tactical recommendations.
1. Automated tactical what-if simulation.
1. Fully automated video clipping from third-party broadcast streams.
1. Autonomous AI tactical decisions.

These features create expensive dependencies and high expectations. Build the ingestion, storage and analytics foundations first.

21. Product success metrics

Activation

1. Percentage of new users who follow at least one club or competition.
1. Percentage of new users who reach a useful match story or tactical explanation without needing a tutorial.
1. Percentage who open their first match centre.
1. Percentage who open or complete their first tactical explanation.
1. Time from sign-up to first useful football insight.

Engagement

1. Weekly active football fans and tactical explorers.
1. Matches explored per active user.
1. Tactical explanations opened, saved or shared.
1. Searches that lead to a match, team or player exploration.
1. Return rate after a match explanation.

- Use of deeper analysis views by users who opt into them.
- Saved concepts and learning progress.

Product quality

- Metric data coverage and freshness by competition.
- Evidence coverage for automated insights.
- Percentage of insights opened for deeper exploration.
- User helpfulness feedback and correction/dismissal rate, segmented by Quick / Explore / Advanced depth.
- AI answer evidence coverage and abstention quality.
- Report generation success.
- App crash-free sessions and ANRs.

Business

- Free-to-paid conversion and churn.
- Monthly recurring revenue.
- Ad revenue per free active user without harming retention.
- Data-provider cost per active user and analysed match.
- Pro feature adoption by audience segment.

North-star metric

Weekly active users who meaningfully explore at least one evidence-backed tactical insight. This is more representative of the product's purpose than raw screen views or app opens.

22. Key risks and mitigations

Risk	Why it matters	Mitigation
Data licensing	A free sample dataset may not permit commercial display, resale or model training	Contract review before production; provider-specific rights registry
Incomplete tracking coverage	Many leagues/matches may not have tracking	Capability flags per match; never imply unavailable data exists
Vendor lock-in	Provider IDs and event schemas differ	Canonical model, adapters, raw payload retention where permitted
Poor tracking quality	Identity swaps, occlusion and smoothing errors can distort metrics	Confidence flags, anomaly detection, smoothing and QA
Misleading metrics	Different vendors use different metric definitions	Version definitions, display methodology and source labels
High data cost	Tracking feeds and video can be expensive	Limit competition scope; meter cost per match and entitlement
AI hallucinations	Fans, analysts or coaches may believe false tactical claims	Structured retrieval, evidence citations, plain-language uncertainty, abstention and audit
Android battery/performance	Tactical visualization and video can be heavy	Server-side analytics, downsampling, efficient rendering and benchmarking
Payment fraud	Client purchase callbacks can be spoofed	Backend verification, RTDN, idempotency and reconciliation
Ad experience damage	Intrusive ads reduce trust and interfere with analysis	Restrained placements; keep tactical workflow and paid screens ad-free
Youth player privacy	Academy data may involve minors	Data minimization, guardian/club permissions, retention and legal review

23. Immediate execution checklist

Before writing production code

1. [ ] Validate the four primary audiences: curious fan, engaged fan, tactical enthusiast and coach/analyst.
1. [ ] Choose the initial competition and country coverage based on data availability and licensing.
1. [ ] Review Sportmonks free/developer access and inspect actual event/stat fields.
1. [ ] Download StatsBomb Open Data and SkillCorner Open Data for development.
1. [ ] Build one event-data parser and one tracking-data parser.
1. [ ] Confirm intended commercial use with every data provider.
1. [ ] Decide whether the first release is post-match only or also needs live updates.
1. [ ] Define Free/Pro entitlements and per-user monthly usage limits.
1. [ ] Design the first-time journey around following clubs and choosing interests, not creating an organization or squad.
1. [ ] Prototype a plain-language insight card and its deeper technical view.
1. [ ] Create the Android design system before building every feature screen.

First engineering tickets

1. Repository and CI/CD setup for Android, API and workers.
2. PostgreSQL schema for users, followed clubs/competitions/players, saved matches and notes.
3. Credential Manager sign-in and backend token/session flow.
4. Canonical competition/team/player/match/event schema and provider adapter interface.
5. Sportmonks ingestion proof of concept.
6. Redis + BullMQ worker skeleton with retry/idempotency.
7. Match list, team page and match centre APIs.
8. Compose Home, Explore, Match Story, Club and Player screens.
9. StatsBomb event parser tests.
10. SkillCorner tracking parser and tactical-pitch prototype.
11. Tactical glossary, Quick / Explore / Advanced explanation pattern and evidence-linked insight cards.

24. Reference documentation

The following official resources were used to shape this plan. Re-check terms, versions, coverage and pricing before procurement or release because providers and platform policies can change.

Football data

1. Sportmonks Football API
1. Sportmonks API documentation
1. StatsBomb Open Data repository and terms
1. SkillCorner Open Data documentation
1. SkillCorner Open Data GitHub repository
1. SkillCorner API documentation
1. Opta Vision
1. Sportradar Soccer API overview
1. Sportradar Soccer API basics
1. API-Football

Android architecture and security

1. Android architecture recommendations
1. Android Credential Manager
1. Play Integrity API
1. OWASP Mobile Application Security

Payments and ads

1. Google Play Billing integration
1. Google Play Billing release notes
1. Google Play payments policy
1. Google Play India alternative billing information
1. Google Play Real-time Developer Notifications
1. AdMob UMP privacy SDK for Android

Final recommendation

Build the product in this order:

Data rights and canonical football data model → Android authentication and football personalisation → fan-first discovery and match centre → simple tactical stories and visual explainers → event-based tactical analytics → licensed tracking ingestion → deeper fan exploration and comparisons → Free/Pro, restrained ads and evidence-grounded AI explanations → optional professional workflows and video.

The product should not make fans feel that they have entered a professional coaching tool. It should make them feel that they have gained a better understanding of football. At the same time, the advanced analytical depth should be real, not simplified away. Use progressive disclosure, plain-language interpretation and evidence-linked drill-downs to serve both needs from one platform.

The most important data decision remains: do not promise full tracking-based tactical analysis based on event APIs alone. Make data coverage visible on every match and unlock advanced spatial analysis only when licensed tracking data is available.